

1. Dataset Provenance and Description

This study uses the *Clothing, Shoes and Jewelry* category from the Amazon Reviews 2023 dataset developed by McAuley Lab [1]. The dataset contains customer review information, including ratings, review titles and text, product identifiers, pseudonymous user identifiers, timestamps, helpful-vote counts, image references, and verified-purchase status [2]. The data are provided in JSON Lines (JSONL) format, where each review is stored as an individual record, making the format suitable for large-scale data processing. The complete Amazon Reviews 2023 collection covers reviews from May 1996 to September 2023, while the subset processed in this study covers December 1999 to March 2023.

The dataset is made available primarily for research purposes, although the maintainers do not specify a separate licence for its use [3]. For this project, the data are therefore used only for non-commercial academic purposes. The original creators are appropriately acknowledged, review-level data are not redistributed, and no attempt is made to identify individual users from their pseudonymous identifiers.

2. Dataset Characteristics and Processing

The complete *Clothing, Shoes and Jewelry* review file was downloaded after its size was verified using the Hugging Face API. The raw and working datasets use JSON Lines (JSONL), where each line represents a single review record. This record-based structure is suitable for large-scale processing because complete records can be read and processed in parallel. To create the processing dataset, complete JSONL records were copied sequentially from the working file until the resulting file exceeded 3 GiB. This ensured that the final record was complete rather than truncated. Dataset sizes are reported in both decimal gigabytes (GB) and binary gibibytes (GiB).

Table 1: Dataset size across processing stages.

Dataset stage	Size and description
Raw dataset	27,810,080,533 bytes (27.810 GB; 25.900 GiB), representing the original publicly available category file.
Working dataset	27,810,080,533 bytes (27.810 GB; 25.900 GiB). The complete category file was downloaded without filtering or reduction.
Processing dataset	3,221,225,482 bytes (3.221 GB; 3.000 GiB), containing 7,231,543 complete JSONL records processed using PySpark 4.0.0.

The processing dataset contains 10 source variables: product and parent-product identifiers, pseudonymous user identifier, rating, review title, review text, timestamp, helpful-vote count, verified-purchase status, and image references. Four additional variables—review time, review year, review-text length, and title length—were derived during data preparation. The resulting DataFrame therefore contains 14 variables. The processed records span 1 December 1999 to 25 March 2023 and include numeric and Boolean attributes, timestamps, identifiers, semi-structured image references, and unstructured review text.

The raw and working datasets have the same size because the complete public file was downloaded without reduction; however, they represent different stages of the data-processing pipeline. The 3.221 GB processing dataset is a sequential prefix of the working file rather than a random sample. Consequently, its temporal and product composition may differ from that of the complete 27.810 GB dataset, which should be considered when interpreting the results.

Data preparation and analysis were performed using PySpark. Ratings, helpful votes, and timestamps were converted to appropriate data types, the four derived variables were created, and the DataFrame was repartitioned into 64 partitions and cached before aggregation. Large-scale transformations and aggregations were performed in Spark, while Pandas was used only to visualise the small aggregated outputs produced by Spark.

3. Data Quality and Preparation

The processing dataset was loaded from JSONL into PySpark and its schema was inspected before analysis. Ratings were converted to numeric values, helpful-vote counts and timestamps to long integers, and millisecond timestamps to readable Spark timestamps. Four variables were then derived: review time, review year, review-text length, and title length. The resulting 14-column DataFrame was repartitioned into 64 partitions and cached for repeated quality checks and aggregations.

Data completeness was assessed by calculating null values across all analysed columns. No null values were identified; however, 495 records (0.0068%) contained empty review text. These records were retained because their other attributes remained valid and they represented only a very small proportion of the dataset. Validity checks also confirmed that all ratings were within the expected 1–5 range and that helpful-vote counts were non-negative.

Potential duplicates were identified using the pseudonymous user identifier, parent-product identifier, timestamp, and review text. This identified 23,544 potential duplicate records (0.3256%). They were flagged rather than removed because matching values do not necessarily indicate invalid records; for example, they may arise from product variants

or changes to product listings. The duplicate check is therefore treated as an indicator of possible duplication rather than definitive evidence of duplicate reviews.

Extreme values were also examined. Eighteen reviews contained more than 10,000 characters, while helpful-vote counts were strongly right-skewed, with a median of 0, mean of 0.919, and maximum of 41,199. These observations were retained because unusually long or highly visible reviews may be genuine. However, the extreme helpful-vote values can influence mean-based statistics, so averages are interpreted together with medians and distributional results.

Several limitations remain. Customer reviews are voluntarily submitted and may therefore be affected by self-selection, platform incentives, product changes, or review manipulation. Pseudonymous identifiers also limit the assessment of demographic representativeness. In addition, the 3.221 GB processing dataset is a sequential prefix rather than a random sample of the complete 27.810 GB file. Its temporal and product composition may therefore differ from the full dataset. Consequently, the reported findings describe the processing subset and should not be interpreted as exact estimates for the complete category dataset.

4. Exploratory Data Analysis

The EDA revealed a strong concentration of positive ratings. Five-star reviews accounted for 64.27% of the 7,231,543 records, followed by four-star reviews at 15.01%, while one-star reviews represented only 6.53%. Consequently, the mean rating was 4.253 and the median was 5, indicating that reported customer experiences were predominantly positive. However, this imbalance means that overall rating statistics are strongly influenced by the large number of five-star reviews.

Review behaviour also differed across rating levels. Review text had an overall mean length of 171 characters and a median of 104, indicating a right-skewed distribution. Three- and four-star reviews were longest on average (approximately 210 and 216 characters), compared with 152 characters for five-star reviews. Helpful-vote counts were even more skewed: the median was zero, while the mean was 0.919 and the maximum was 41,199. One-star reviews received the highest mean number of helpful votes (1.325), suggesting that negative experiences may attract greater engagement. These relationships are descriptive associations and should not be interpreted as causal.

Review activity increased substantially over time, from 106,455 records in 2013 to a peak of 1,133,095 in 2021, followed by 1,054,750 in 2022. The 237,486 reviews observed in 2023 do not represent a full-year decline because the processing dataset ends on 25 March 2023. Mean annual ratings remained relatively stable during years with substantial review volumes, while results from the earliest years should be interpreted cautiously because of their small sample sizes.

Verified purchases represented 92.48% of the processing dataset. Their mean rating (4.255) was only slightly higher than that of unverified reviews (4.228), a difference of approximately 0.027 stars and therefore small in practical terms. Product activity was considerably more uneven: the dataset contained 1,836,721 distinct parent products, with a median of only one review per product, compared with 37,409 reviews for the most-reviewed product. This indicates a strongly long-tailed distribution in which a small number of products account for disproportionately high review activity.

Overall, the EDA identifies rating imbalance, skewed review engagement, temporal changes, differences in review behaviour across rating levels, and strong product-level concentration. Supporting statistics and visualisations are provided in Appendix A. These findings describe the 3.221 GB sequential processing subset and may not generalise exactly to the complete category dataset.

5. The 7 Vs of Big Data

Table 2: Application of the 7 Vs of Big Data to the processing dataset.

V	Evidence from the dataset	Relevance
Volume	The complete category file is 27.810 GB, while the 3.221 GB processing subset contains 7,231,543 records and 14 analysed variables.	This scale motivates distributed loading, partitioning, caching, and aggregation with PySpark rather than relying on in-memory analysis.
Velocity	Reviews accumulated over time, with annual volume exceeding one million records from 2019 to 2022 and reaching 1,133,095 in 2021.	Review activity changes over time, making temporal analysis important. Here, velocity reflects historical data accumulation rather than real-time ingestion.
Variety	The data combine unstructured review text, numeric ratings and votes, Boolean purchase verification, timestamps, identifiers, and semi-structured image references.	These different data types require schema inspection, type conversion, timestamp processing, and text-derived variables.
Veracity	Quality checks identified 495 empty-text records, 23,544 potential duplicates (0.3256%), 18 unusually long reviews, and helpful votes ranging up to 41,199.	Potential duplicates, extreme values, and unreliable reviews can distort summary statistics and product comparisons, requiring cautious interpretation.
Value	The data reveal rating patterns, review-volume trends, helpfulness, verified-purchase behaviour, review-length differences, and highly reviewed products.	These insights can support product-quality monitoring, merchandising, review ranking, and customer-experience decisions.
Variability	Review behaviour differs across time, rating levels, and products. The median product has one review, while the most-reviewed product has 37,409.	Such variation means that overall averages can conceal temporal changes and substantial differences in product-level activity.
Visualization	Six Spark-derived figures summarise ratings, annual activity, mean ratings, verified purchases, review behaviour, and product concentration.	Visualisation converts millions of records into interpretable patterns; only small aggregated Spark outputs are transferred to Pandas for plotting.

6. Business/Societal Value and Limitations

The analysis provides business value by helping retailers and manufacturers identify products with high review volumes and unusually low ratings, monitor customer experience over time, and prioritise products that may require further

investigation. Patterns in ratings, review volume, verified purchases, and helpful votes may also support merchandising, quality-control, and review-ranking decisions. From a societal perspective, aggregated review information can improve product transparency and assist consumers in making more informed purchasing decisions.

These applications should, however, be interpreted within the limitations of the data. First, the analysis is observational, so differences between groups, such as verified and unverified purchases, represent associations rather than causal effects. Second, reviews are voluntarily submitted and may therefore over-represent customers with particularly positive or negative experiences. Third, pseudonymous identifiers provide no demographic information, making it impossible to assess whether reviewers are representative of the broader customer population.

The findings should therefore be treated as evidence from the processing subset rather than as definitive estimates for the complete category dataset. Before being used for operational decisions, they should be validated against the full dataset and, where possible, additional business evidence. Ethical use of the data requires aggregate reporting, protection of pseudonymous identifiers, and avoidance of attempts to identify individual reviewers.

Reproducibility: Code, notebooks, figures, and supporting outputs are available in the [GitHub repository](#). Raw review files are excluded because of their size.

References

- [1] Hou et al. (2024), *Bridging Language and Items for Retrieval and Recommendation*. <https://arxiv.org/abs/2403.03952>
- [2] McAuley Lab, *Amazon Reviews 2023 – Hugging Face*. <https://huggingface.co/datasets/McAuley-Lab/Amazon-Reviews-2023>
- [3] McAuley Lab, *Dataset licence discussion*. <https://huggingface.co/datasets/McAuley-Lab/Amazon-Reviews-2023/discussions/1>

A. Appendix

A.1 Exploratory Visualisations

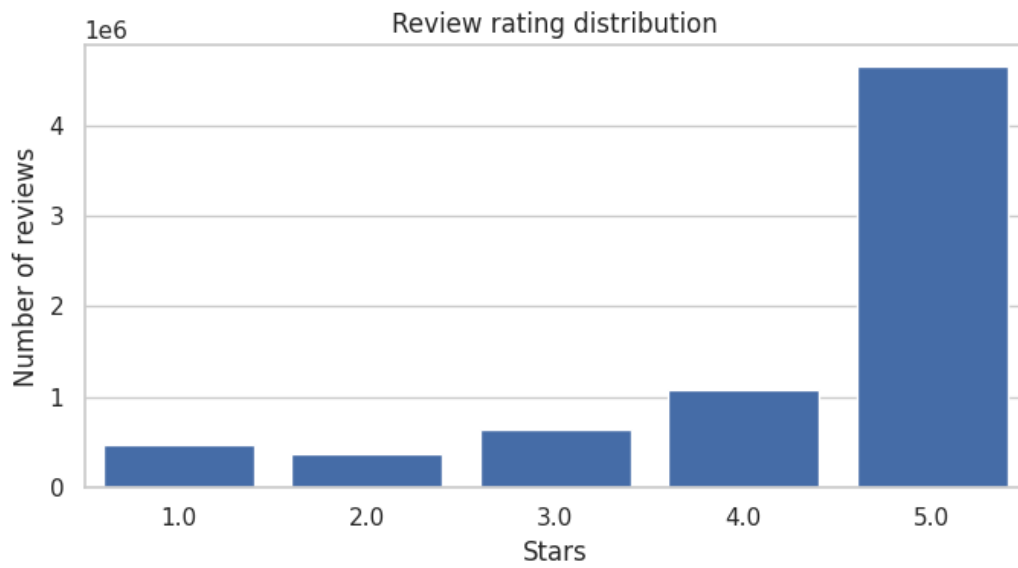


Figure 1: Rating distribution for the 7,231,543-record processing subset.

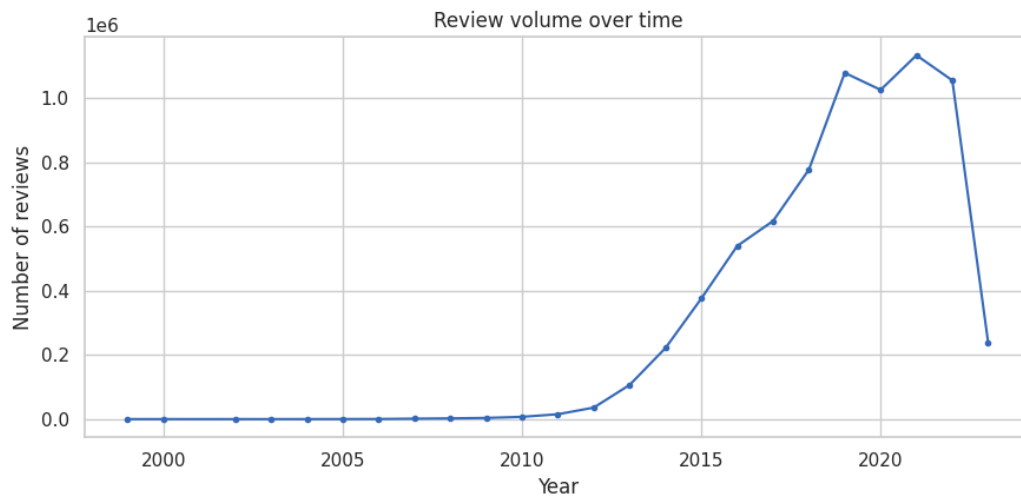


Figure 2: Annual review volume. The 2023 observations cover only the period up to 25 March and therefore do not represent a complete year.

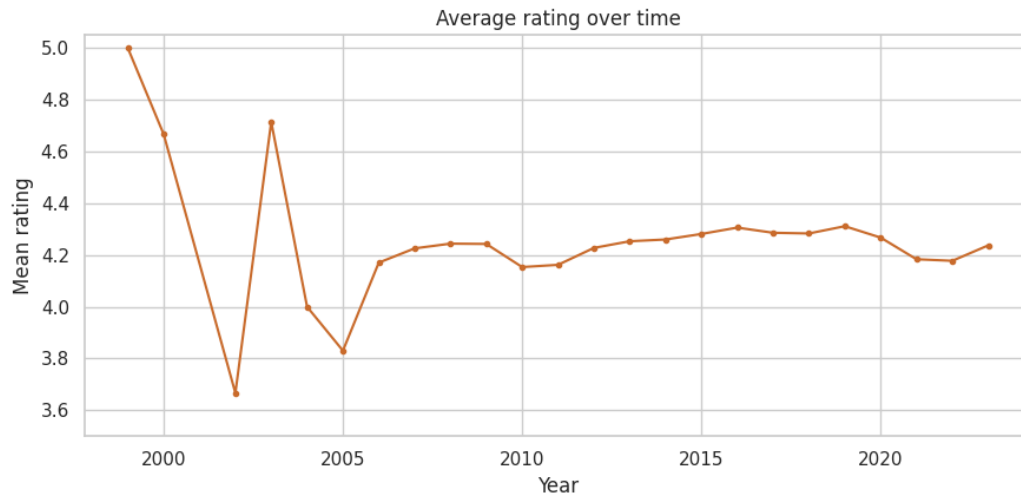


Figure 3: Mean rating by review year. Early years contain relatively few observations and should therefore be interpreted cautiously.

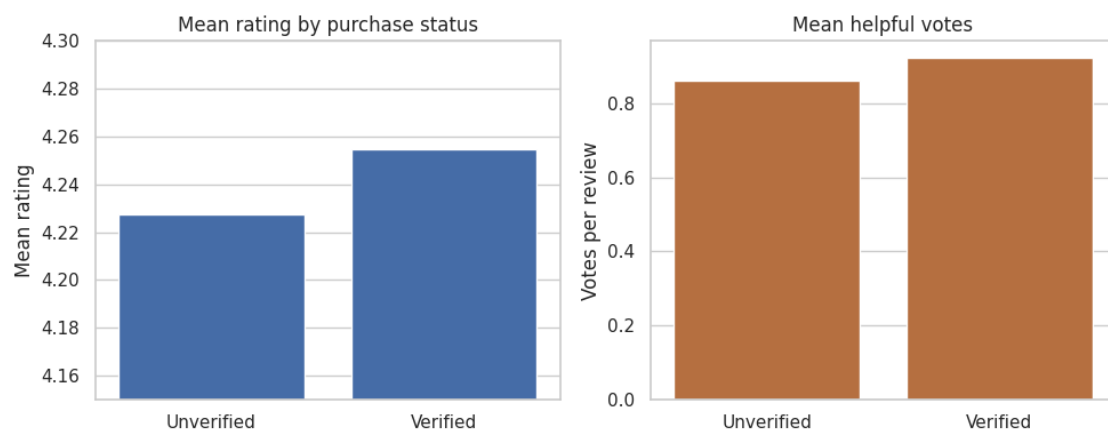


Figure 4: Mean rating and mean helpful votes by verified-purchase status.

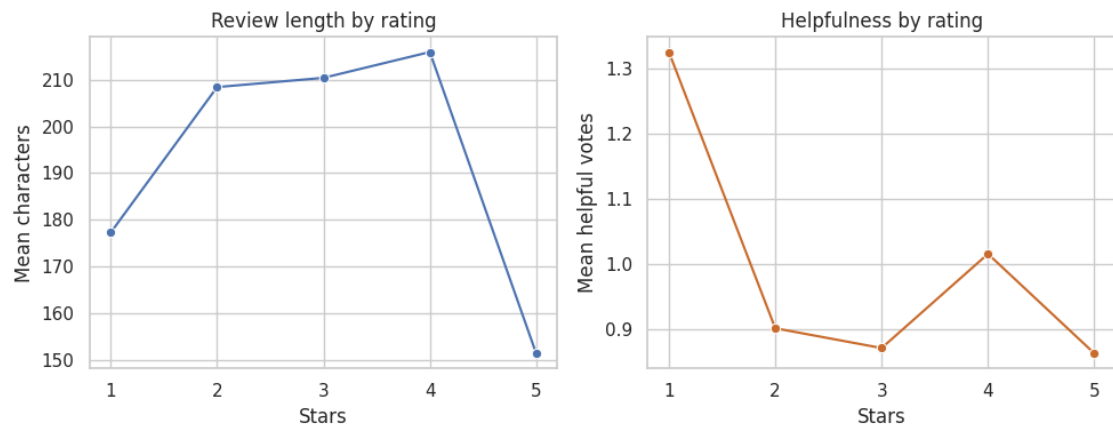


Figure 5: Mean review-text length and helpful votes across rating levels.

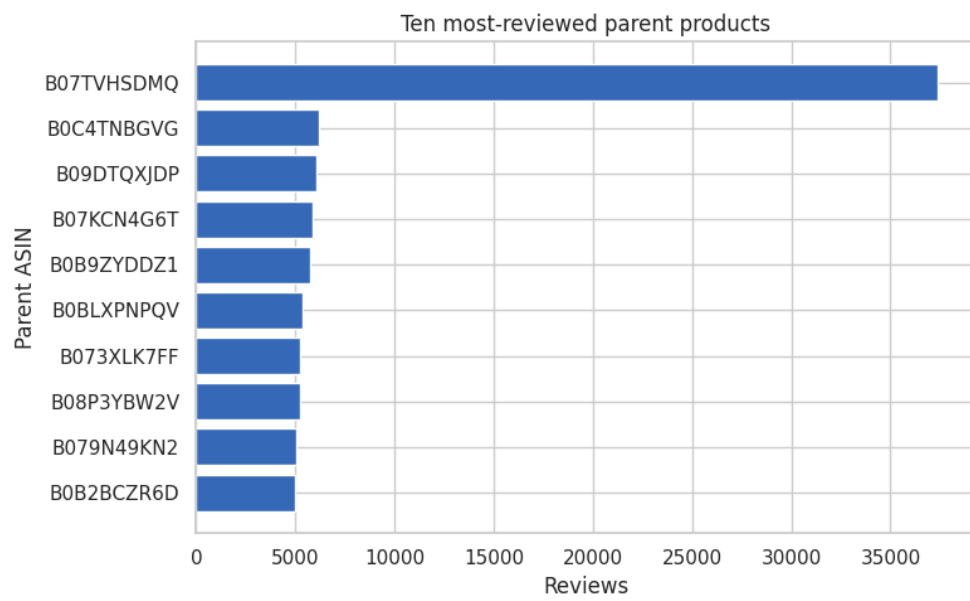


Figure 6: Ten most-reviewed parent products in the processing subset.

A.2 Supporting Result Tables

Table 3: Descriptive statistics for the processing subset.

Statistic	Rating	Helpful votes	Text length
Count	7,231,543	7,231,543	7,231,543
Mean	4.253	0.919	171.079
Std. deviation	1.210	17.696	225.653
Minimum	1	0	0
25th percentile	4	0	45
Median	5	0	104
75th percentile	5	0	213
Maximum	5	41,199	31,254

Table 4: Rating distribution and review characteristics by rating level.

Stars	Reviews	Share (%)	Mean length	Median length	Mean helpful
1	472,381	6.53	177.3	118	1.325
2	376,802	5.21	208.4	142	0.901
3	649,147	8.98	210.4	139	0.871
4	1,085,590	15.01	215.9	135	1.016
5	4,647,623	64.27	151.5	89	0.863

Table 5: Comparison by verified-purchase status.

Status	Reviews	Mean rating	Mean helpful votes
Unverified	543,312	4.228	0.863
Verified	6,688,231	4.255	0.923

Table 6: Data-quality and product-concentration results.

Measure	Result
Null values across analysed columns	0
Empty review text	495 (0.0068%)
Text longer than 10,000 characters	18
Potential duplicate rows	23,544 (0.3256%)
Distinct parent products	1,836,721
Median reviews per product	1
99th percentile reviews per product	48
Largest product review count	37,409

Table 7: Annual review counts and mean ratings, 2010–2023.

Year	Reviews	Mean rating
2010	7,085	4.154
2011	15,169	4.162
2012	35,864	4.228
2013	106,455	4.253
2014	221,415	4.260
2015	375,728	4.282
2016	539,145	4.307
2017	616,925	4.286
2018	777,115	4.283
2019	1,078,297	4.311
2020	1,025,346	4.267
2021	1,133,095	4.184
2022	1,054,750	4.178
2023	237,486	4.237